Ericsson PE Node - Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) Resolution SOP

Node: JAY-PE-09 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_5259

Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on the Ericsson PE Node (Role: PE, Node ID: JAY-PE-09, Site Location: Jayanagar Metro Site) is a critical failure domain that can be attributed to the underlying hardware and protocol mechanics of the Optical Transport Network (OTN). The OTN is responsible for transmitting high-speed data signals over optical fibers, and any degradation or loss of signal strength can result in a critical failure. In this specific scenario, the failure is likely caused by either a faulty SFP/XFP, a fiber cut, or a power loss issue in the receive or transmit path. The Ericsson PE Node, being a critical component of the OTN, is responsible for switching and routing optical signals, and any failure in this node can have a ripple effect on the entire network. Therefore, it is essential to troubleshoot and resolve this failure domain promptly to minimize network downtime and ensure the continued operation of critical services.

Pre-requisite Checks

- Check: Verify that the Jayanagar Metro Site is accessible and that remote hands are dispatched to the site.

- Check: Confirm that the redundant link is functioning properly and that the network is not in a single-point-of-failure scenario.

Potential Root Causes

Cause	Probability
Faulty SFP/XFP	High
Fiber cut	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show tech:otn::status, Expected Output: OTN status: Normal, Description: This command will display the current status of the OTN, including the SFP/XFP status, fiber status, and Rx/Tx power levels. The output will help identify if there are any issues with the SFP/XFP or fiber cuts.

Command: show tech:otn::sfp::status, Expected Output: SFP/XFP status: OK, Description: This command will display the status of the SFP/XFP, including the temperature, voltage, and signal strength. The output will help identify if the SFP/XFP is faulty.
```

```
Command: show tech:otn::fiber::status, Expected Output: Fiber status: OK, Description: This command will display the status of the fiber, including the transmit and receive power levels. The output will help identify if there are any issues with the fiber cuts or power loss.
```

Resolution Steps

Step 1: Step: Replace faulty SFP/XFP, Command: replace sfp/xfp , Description: Replace the faulty SFP/XFP with a new one. Make sure to select the correct slot number and SFP/XFP type.

Step 2: Step: Verify fiber status, Command: show tech:otn::fiber::status, Description: Verify that the fiber status is OK after replacing the SFP/XFP. If the fiber status is still not OK, proceed to the next step.

Step 3: Step: Verify Rx/Tx power levels, Command: show tech:otn::status, Description: Verify that the Rx/Tx power levels are within the normal range after replacing the SFP/XFP and verifying the fiber status. If the power levels are still not within the normal range, proceed to the next step.

Post-Check Verification

```
Command: show tech:otn::status, Expected Output: OTN status: Normal, Description: Verify that the OTN status is Normal after completing the resolution steps.

Command: show tech:otn::sfp::status, Expected Output: SFP/XFP status: OK, Description: Verify that the SFP/XFP status is OK after completing the resolution steps.

Command: show tech:otn::fiber::status, Expected Output: Fiber status: OK, Description: Verify that the fiber status is OK after completing the resolution steps.
```

Roll-back Procedures

- Step: Restore original SFP/XFP, Command: restore sfp/xfp , Description: Restore the original SFP/XFP if the replacement SFP/XFP fails to resolve the issue.

- Step: Verify OTN status, Command: show tech:otn::status, Description: Verify that the OTN status returns to Normal after restoring the original SFP/XFP.